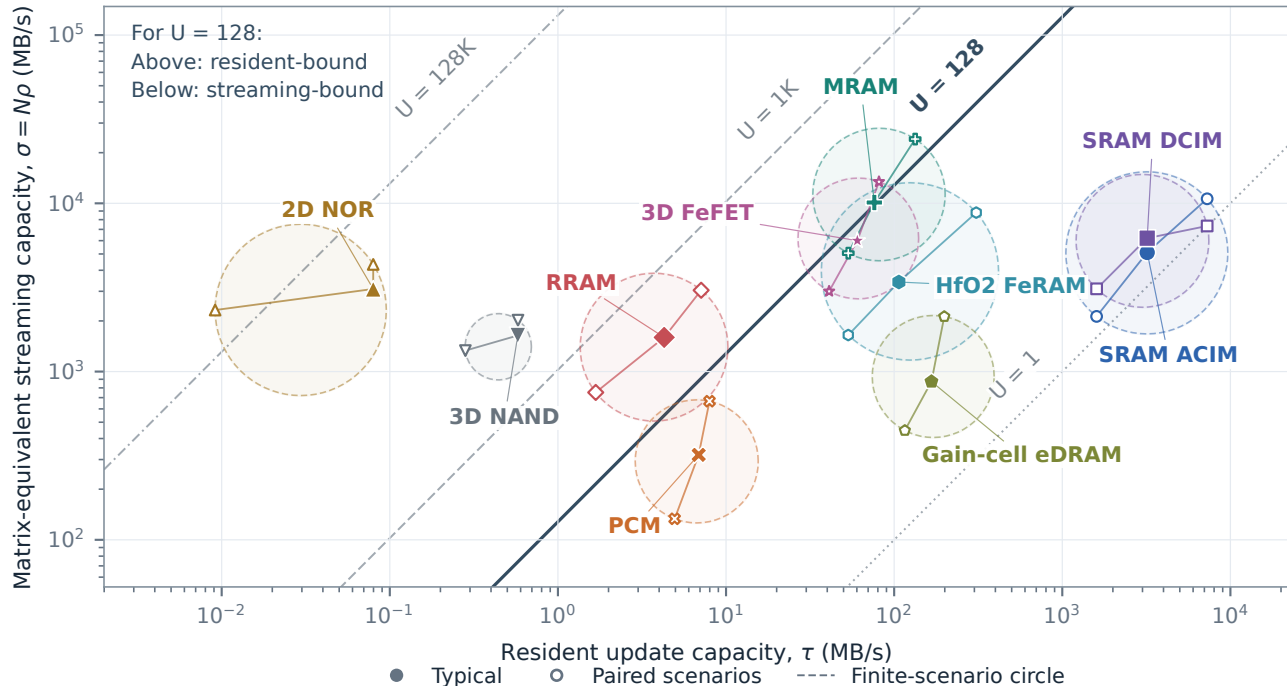


A | Native full-load reuse reference

Each native matrix is loaded once; balance at $\sigma = U\tau$



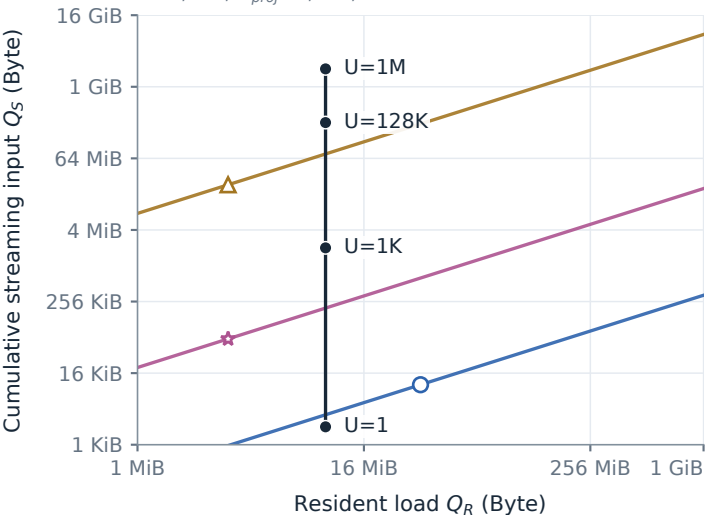
U : served vectors per full load 1K = 1024 MB/s = 10^6 Byte/s

B | QKV demand and native-tile balance

Logical traffic; each 128×128 tile loaded once, serves U vectors

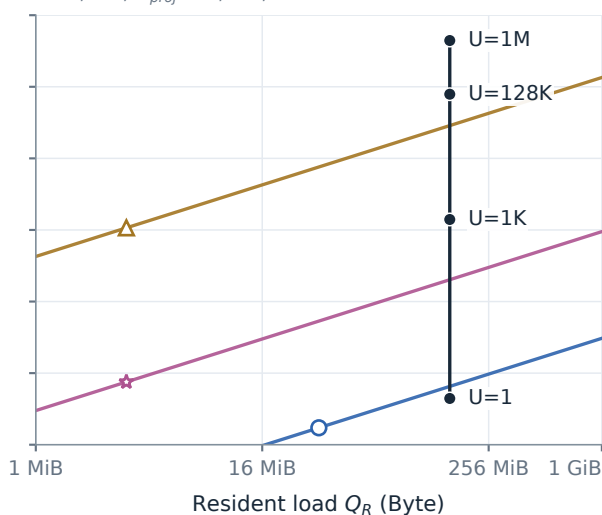
(a) Qwen3.5-2B

$D=2,048$; $N_{proj}=5,120$; load=10 MiB



(b) MiMo-V2.5-Pro

$D=6,144$; $N_{proj}=27,136$; load=159 MiB

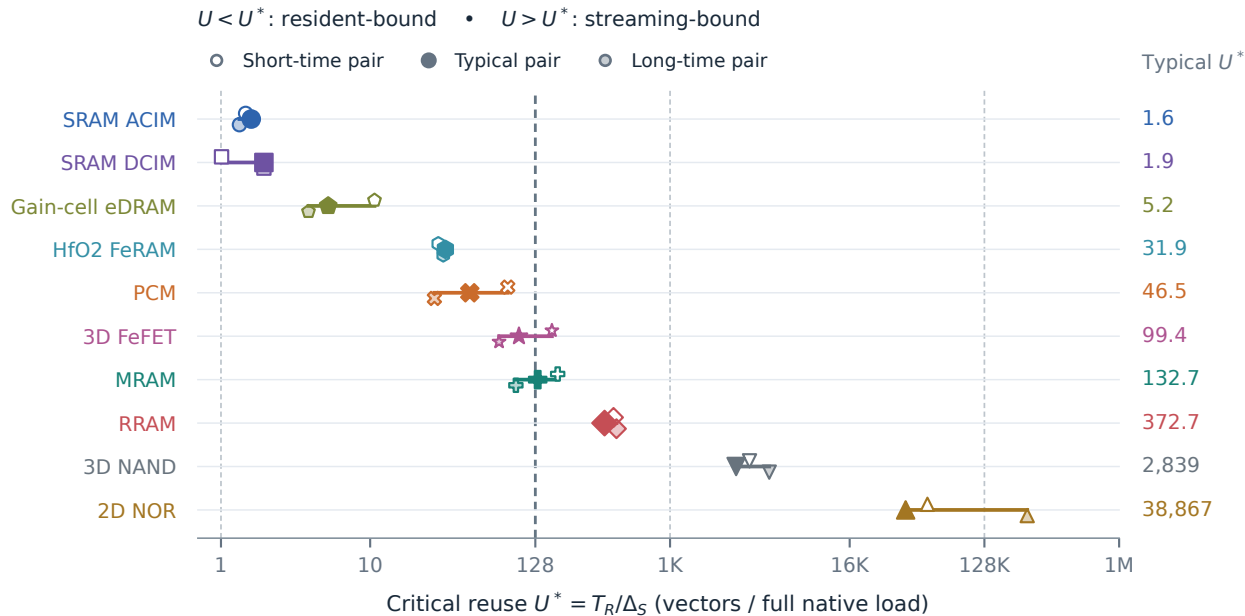


—○— SRAM ACIM ($U^* = 1.60$) —☆— 3D FeFET ($U^* = 99.4$) —△— 2D NOR ($U^* = 38.0K$)

Typical hardware references • ray: $Q_S = Rl_{eff}^* Q_R$

Above each ray: streaming-bound | Below each ray: resident-bound

C | Reuse needed to balance one full load



Dashed lines: Table II(a) reuse references $U = 1, 128, 1K, 128K$

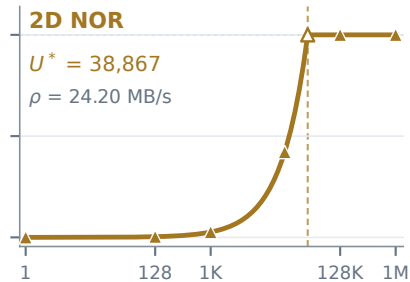
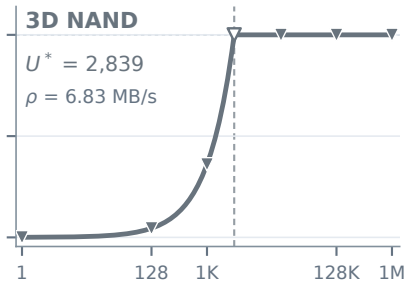
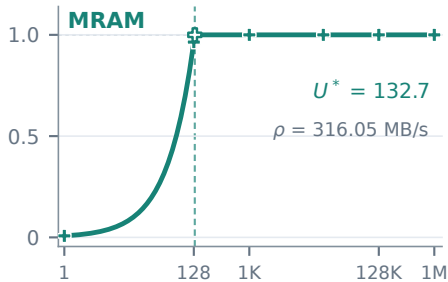
MRAM at $U = 128$: short / typical / long = resident / resident / streaming

D | How reuse approaches each streaming ceiling

Typical paired scenarios: $P_{\text{bound}}/\rho = \min(1, U/U^*)$

Filled: Table II(a) U scan · Open: balance U^*

Reference upper bound P_{bound}/ρ



Reuse U (vectors / full native load)

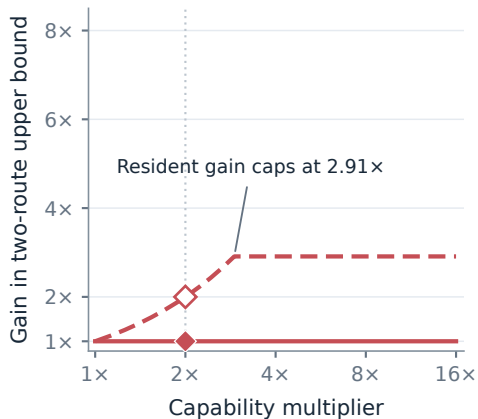
Equal normalized plateaus have different absolute streaming ceilings.

E | Improve the route that limits the bound

Analytic capability scaling at three native full-load service states

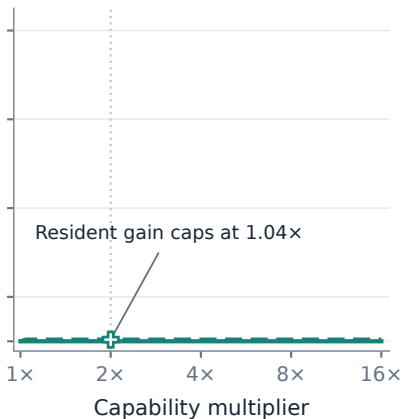
(a) RRAM, U=128

Resident-bound



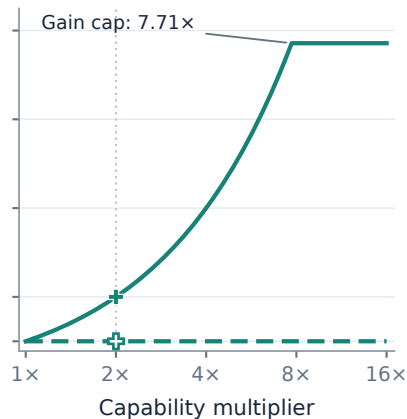
(b) MRAM, U=128

Near balance (resident-bound)



(c) MRAM, U=1K

Streaming-bound



— Streaming only: $\rho \rightarrow f\rho$ - - - Resident only: $\tau \rightarrow f\tau$

Single-route upper-bound gains; capacity factors do not represent equal cost